Case Study 3: Customer Segmentation

Aim: group retailer customers using Recency, Frequency and Monetary value; compare K-Means and DBSCAN, reduce dimensions with PCA, visualize with t-SNE, and propose actionable personas.

Source: D. Chen (2015), Online Retail, UCI Machine Learning Repository. DOI: 10.24432/C5BW33. License: CC BY 4.0.
https://archive.ics.uci.edu/dataset/352/online+retail

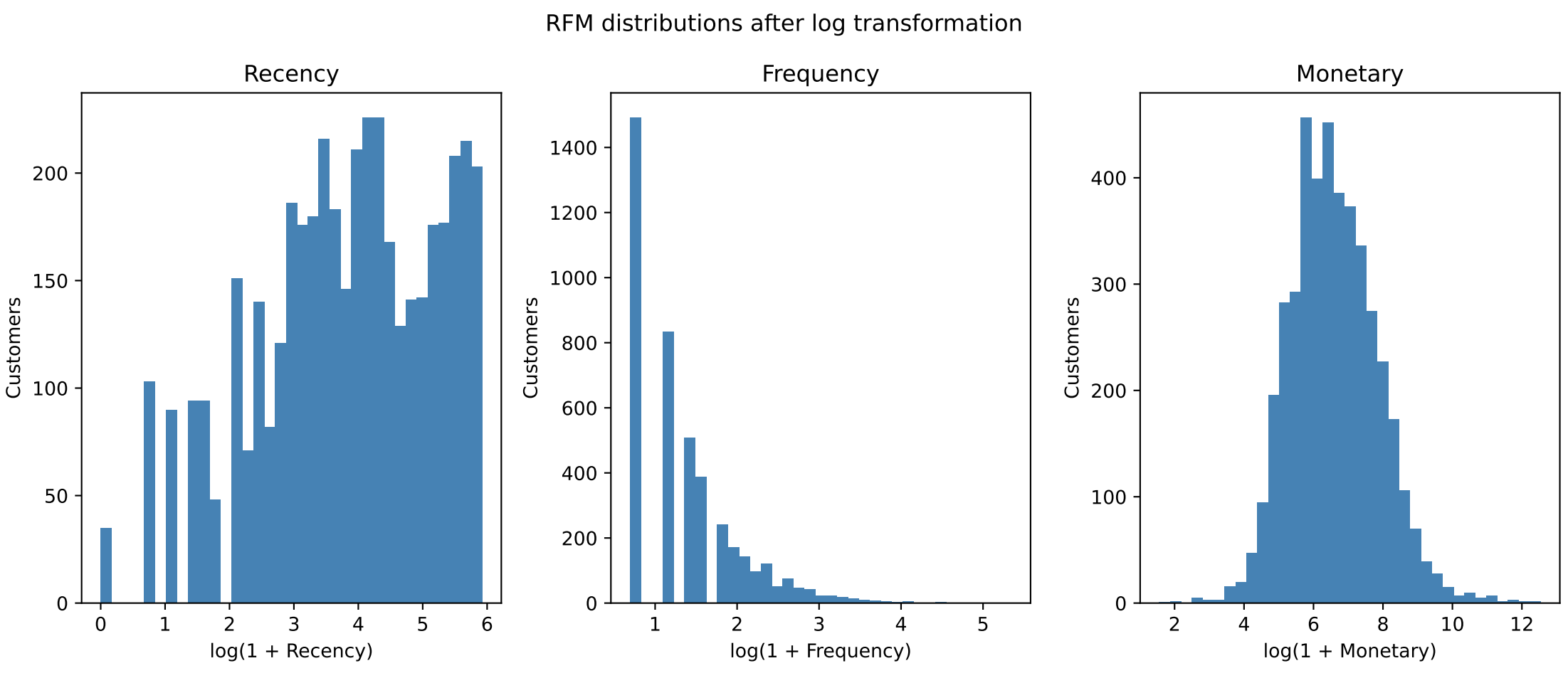
Included data: 541,909 original transaction lines, five RFM columns. Dates: 2010-12-01 to 2011-12-09. No customer subsampling; all valid identified customers are included.

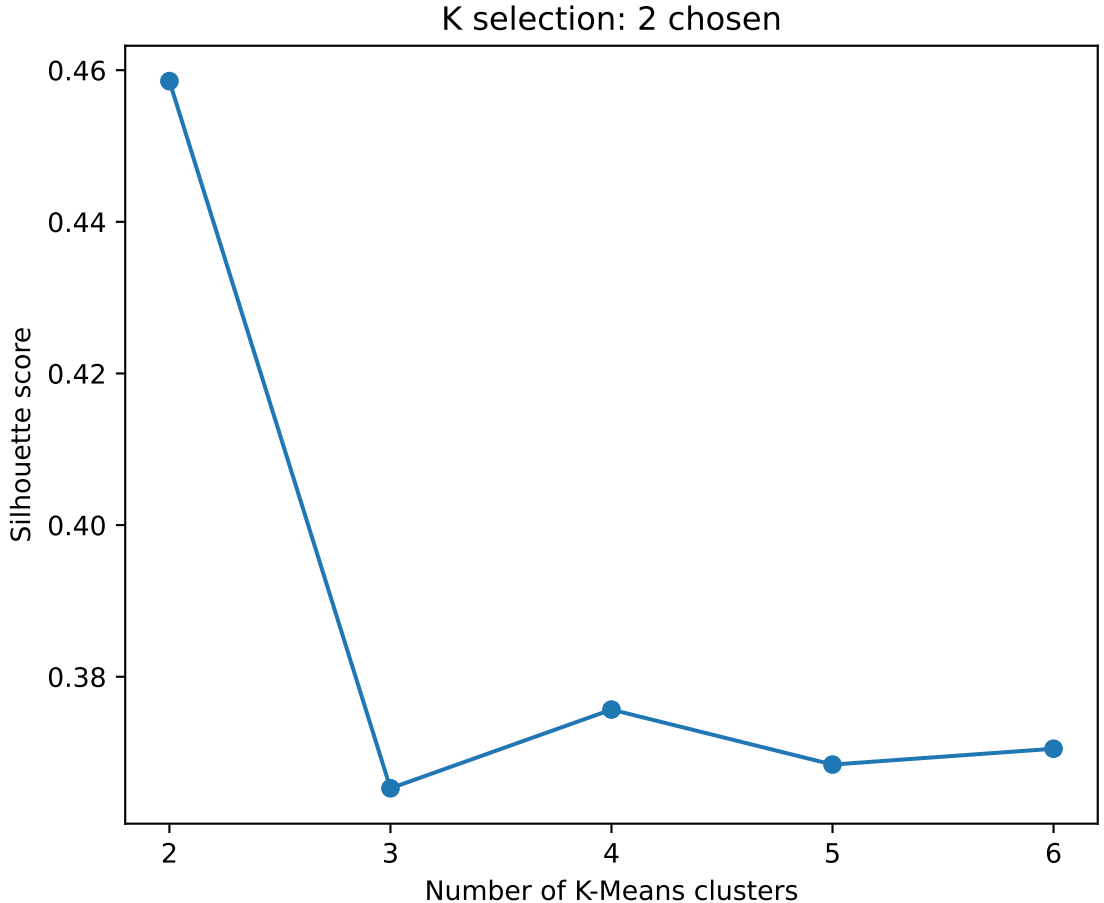
Cleaning retained 397,884 rows and 4,338 customers; removed 144,025 rows with missing required values, canceled invoices, or nonpositive quantity/price. Original line items are preserved, including repeated purchases. Refunds are excluded: Monetary represents gross valid purchase value, not net revenue.

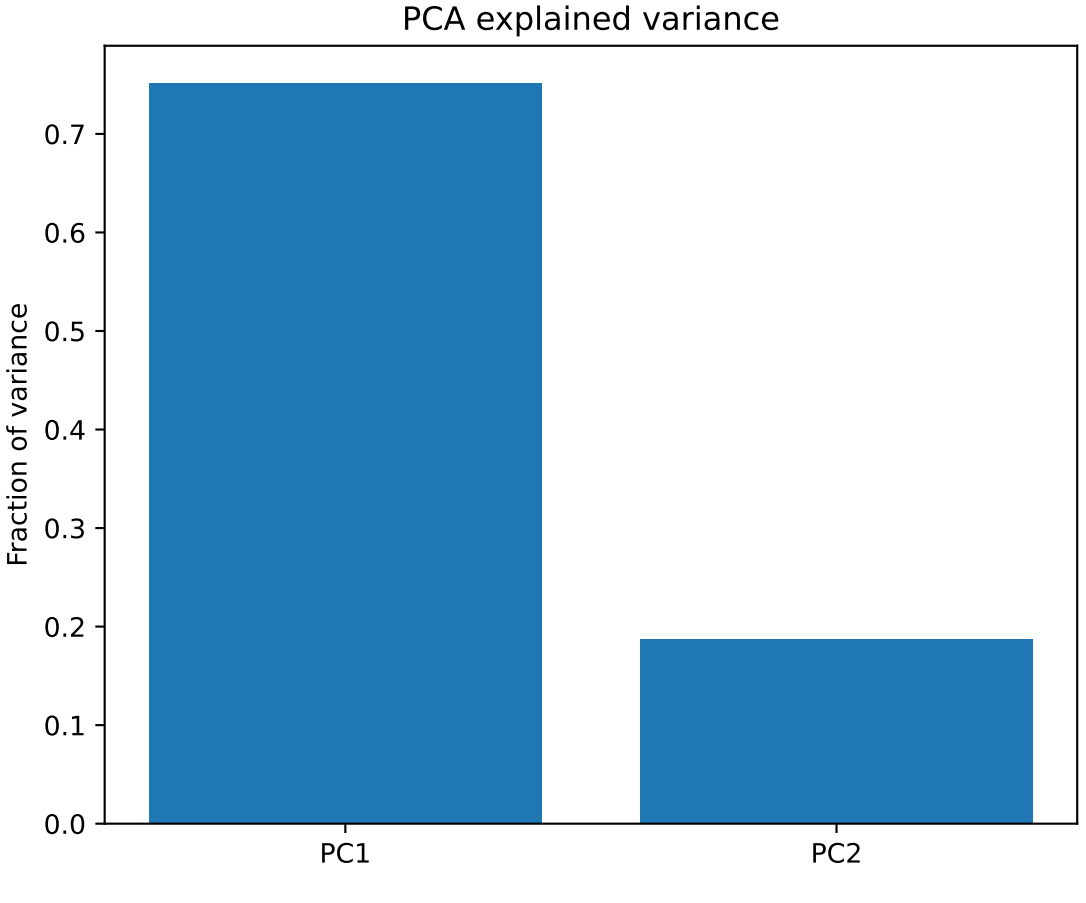
Snapshot: 2011-12-10. Recency = whole days since the latest purchase; Frequency = distinct invoice count; Monetary = sum(Quantity * UnitPrice), GBP.

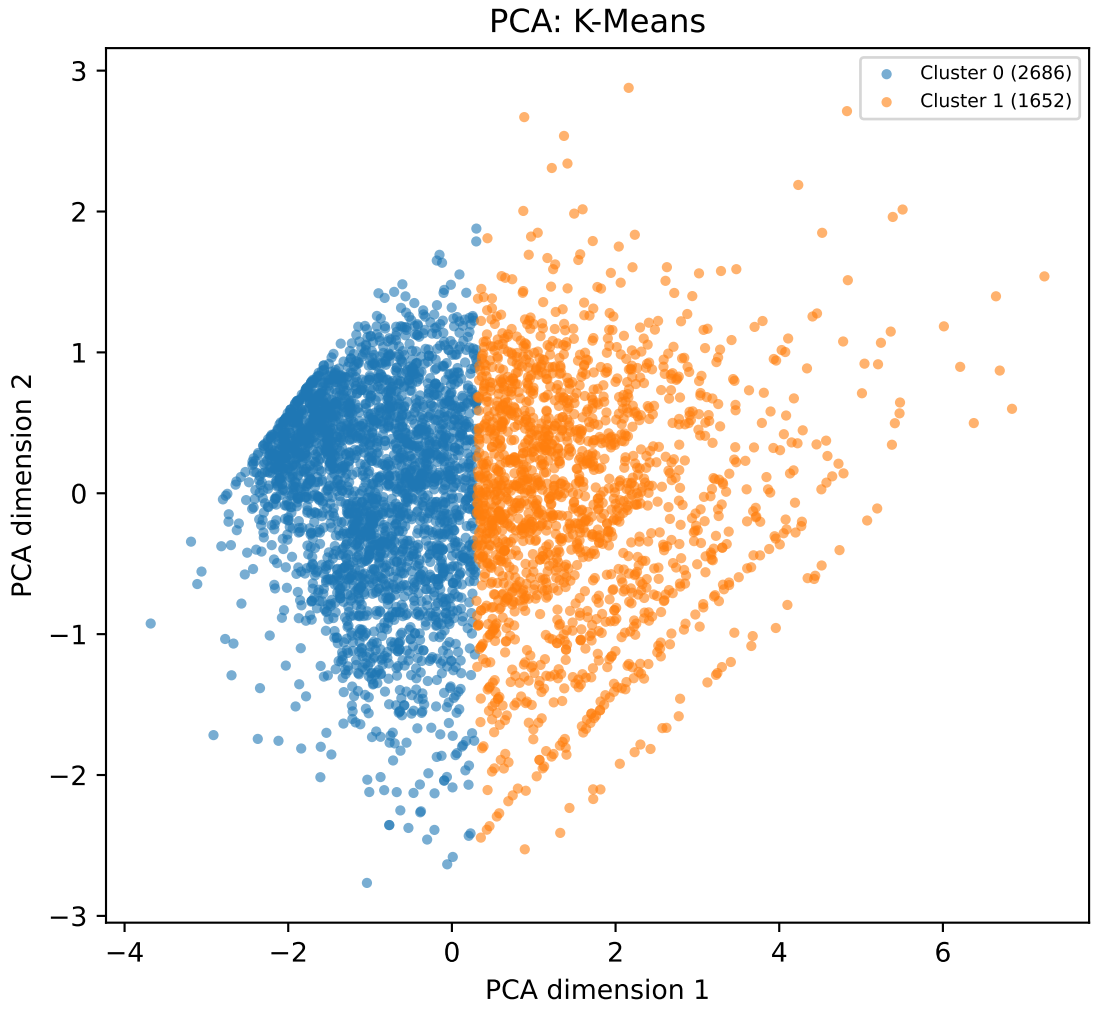
Workflow: clean -> RFM -> log1p -> standardize -> PCA (2D) -> both clustering algorithms. t-SNE uses the standardized three RFM features only for plots.

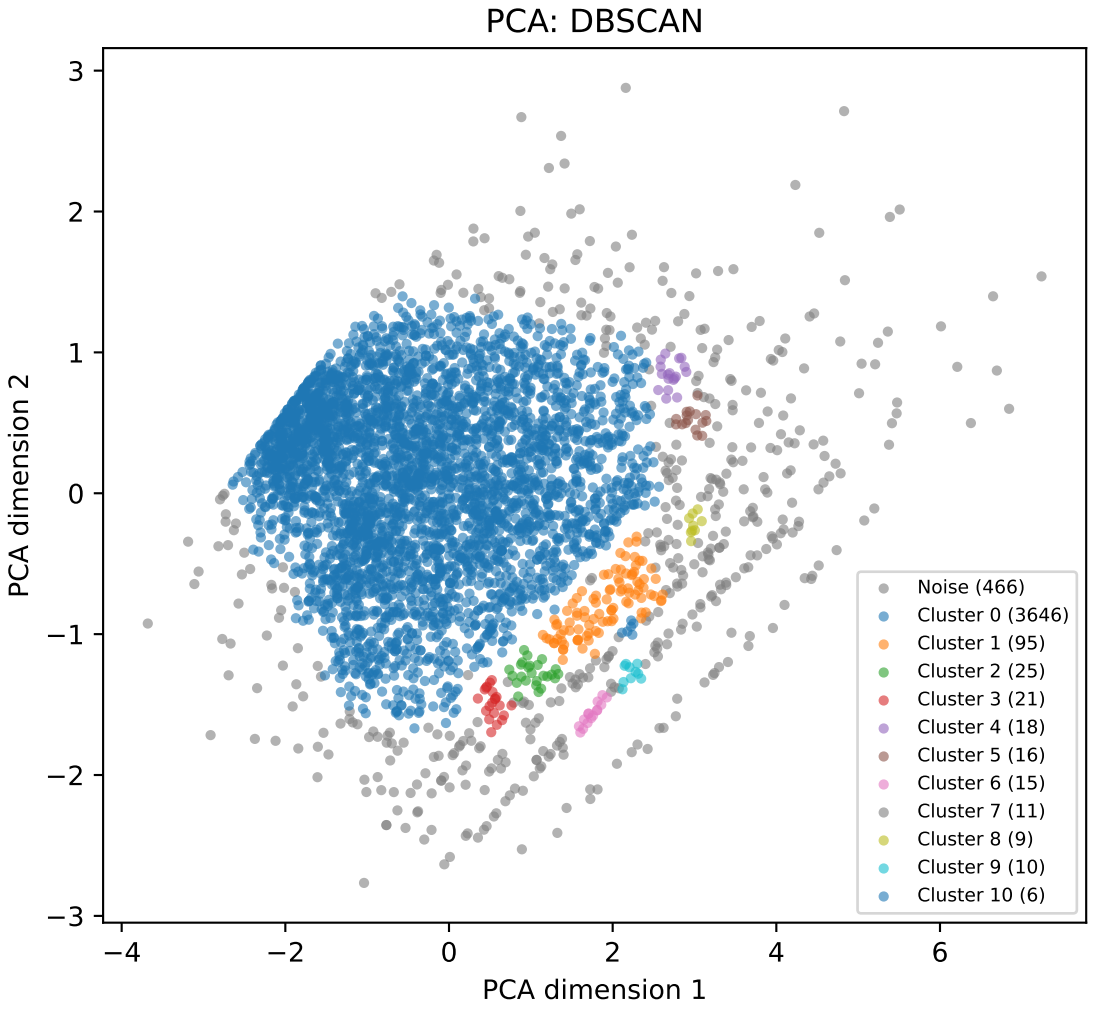
Run: pip install pandas numpy scikit-learn matplotlib; python code.py. Seed: 42. pandas 3.1.0; scikit-learn 1.9.1; NumPy 2.5.4; Matplotlib 3.11.2.

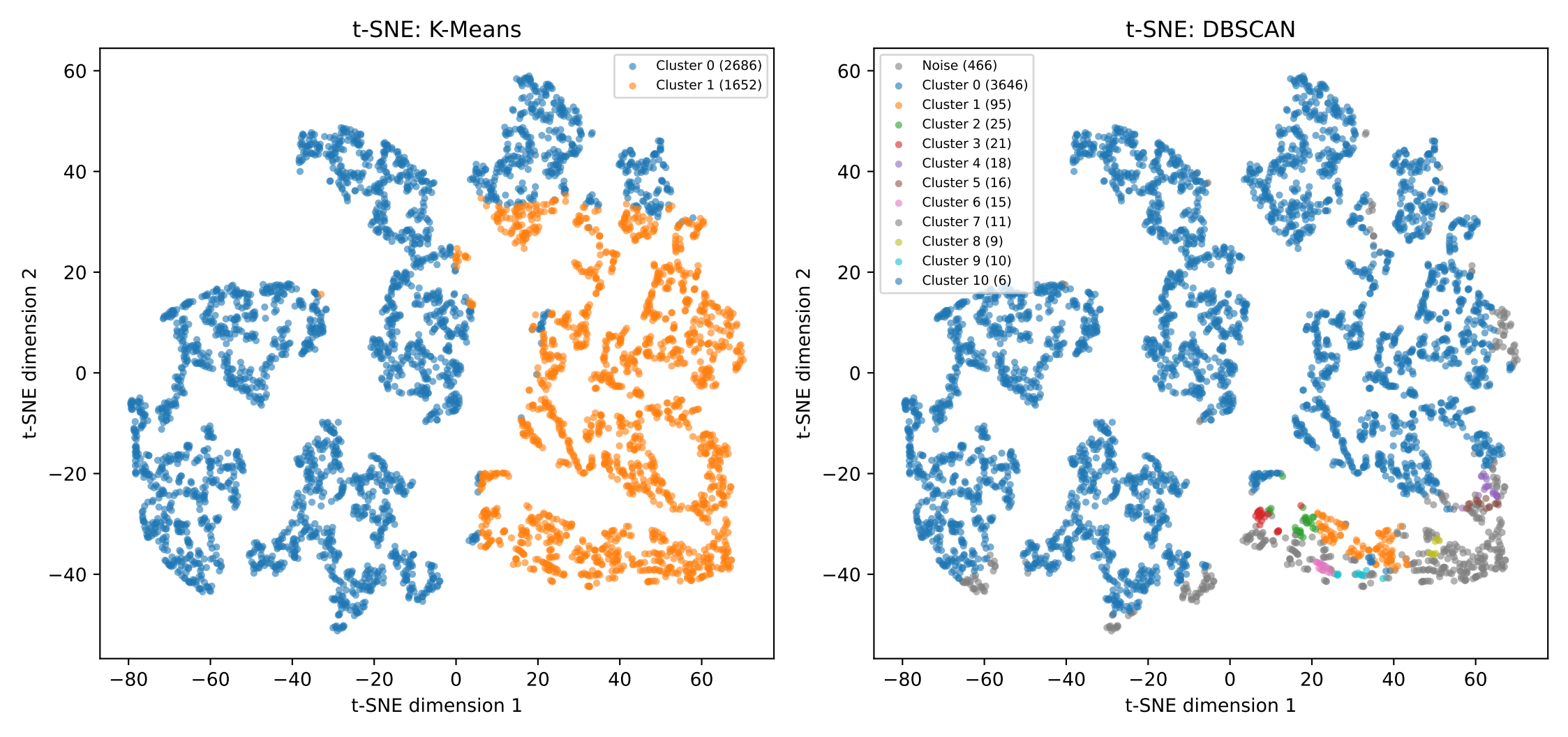

RFM distributions after log transformation
Recency
Frequency
Monetary
Customers
log(1 + Recency)
log(1 + Frequency)
log(1 + Monetary)


K selection: 2 chosen
Silhouette score
0.46
0.44
0.42
0.40
0.38
2
3
4
5
6
Number of K-Means clusters


PCA explained variance
Fraction of variance
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
PC1
PC2


PCA: K-Means
Cluster 0 (2686)
Cluster 1 (1652)
PCA dimension 2
PCA dimension 1
3
2
1
0
−1
−2
−3
−4
−2
0
2
4
6


PCA: DBSCAN
Noise (466)
Cluster 0 (3646)
Cluster 1 (95)
Cluster 2 (25)
Cluster 3 (21)
Cluster 4 (18)
Cluster 5 (16)
Cluster 6 (15)
Cluster 7 (11)
Cluster 8 (9)
Cluster 9 (10)
Cluster 10 (6)
PCA dimension 2
PCA dimension 1
3
2
1
0
−1
−2
−3
−4
−2
0
2
4
6


t-SNE: K-Means
Cluster 0 (2686)
Cluster 1 (1652)
t-SNE dimension 1
t-SNE dimension 2
t-SNE: DBSCAN
Noise (466)
Cluster 0 (3646)
Cluster 1 (95)
Cluster 2 (25)
Cluster 3 (21)
Cluster 4 (18)
Cluster 5 (16)
Cluster 6 (15)
Cluster 7 (11)
Cluster 8 (9)
Cluster 9 (10)
Cluster 10 (6)
t-SNE dimension 1
t-SNE dimension 2

Comparison and limitations

PCA retains 93.9% of standardized log-RFM variance. Clusters and silhouette scores use these two PCs.

K-Means: K=2; silhouette=0.459. Every customer receives a cluster. K-Means favors compact groups and is sensitive to scaling, extreme customers and the chosen K.

DBSCAN: eps=0.15, min_samples=10; clusters=11; noise=10.7%; non-noise silhouette=0.109. Noise is excluded from its silhouette, so that score cannot be compared directly with K-Means without considering coverage. NaN means fewer than two usable clusters. Noise means unusual purchase behavior, not fraud.

K-Means labels evaluated in all three standardized RFM dimensions: silhouette=0.427. DBSCAN non-noise labels in three dimensions: silhouette=0.095. This checks whether separation also appears outside the 2D projection.

t-SNE: perplexity=30, seed=42. Nearby points can reveal local structure; global distances and apparent island sizes are not reliable evidence of a segment. Neither model is fitted on t-SNE coordinates.

Limitations: one historical retailer, gross spend, no demographic or profitability data. K selection uses a fixed silhouette sample of up to 2,000 customers. DBSCAN uses a fixed radius; sensitivity is shown below. Personas are median-based hypotheses, not measured campaign results. Validate actions using consented A/B tests and monitor segments over time.

Methods: https://scikit-learn.org/stable/modules/clustering.html; https://scikit-learn.org/stable/modules/decomposition.html#pca; https://scikit-learn.org/stable/modules/manifold.html#t-sne

KMeans: personas and marketing actions

Cluster 0: Occasional customers. Customers: 2,686 (61.9%). Median R=95 days, F=1 orders, M=GBP 364.76. Action: Test relevant cross-sells and replenishment reminders.

Cluster 1: Loyal high-value customers. Customers: 1,652 (38.1%). Median R=16 days, F=6 orders, M=GBP 2,086.48. Action: Offer early access and loyalty benefits.

DBSCAN: personas and marketing actions

Cluster -1: Unusual customers / noise. Customers: 466 (10.7%). Median R=4 days, F=8 orders, M=GBP 3,504.86. Action: Inspect individually before sending mass campaigns.

Cluster 0: Occasional customers. Customers: 3,646 (84.0%). Median R=63 days, F=2 orders, M=GBP 589.00. Action: Test relevant cross-sells and replenishment reminders.

Cluster 1: Loyal high-value customers. Customers: 95 (2.2%). Median R=4 days, F=6 orders, M=GBP 2,031.78. Action: Offer early access and loyalty benefits.

Cluster 2: Occasional customers. Customers: 25 (0.6%). Median R=4 days, F=4 orders, M=GBP 787.85. Action: Test relevant cross-sells and replenishment reminders.

Cluster 3: Recent occasional buyers. Customers: 21 (0.5%). Median R=4 days, F=2 orders, M=GBP 610.52. Action: Send onboarding tips and a second-purchase offer.

Cluster 4: Loyal high-value customers. Customers: 18 (0.4%). Median R=18 days, F=16 orders, M=GBP 7,365.20. Action: Offer early access and loyalty benefits.

DBSCAN: personas and marketing actions

Cluster 5: Loyal high-value customers. Customers: 16 (0.4%). Median R=10 days, F=16 orders, M=GBP 6,746.95. Action: Offer early access and loyalty benefits.

Cluster 6: Occasional customers. Customers: 15 (0.3%). Median R=1 days, F=5 orders, M=GBP 1,250.76. Action: Test relevant cross-sells and replenishment reminders.

Cluster 7: Loyal high-value customers. Customers: 11 (0.3%). Median R=2 days, F=15 orders, M=GBP 4,654.08. Action: Offer early access and loyalty benefits.

Cluster 8: Loyal high-value customers. Customers: 9 (0.2%). Median R=3 days, F=13 orders, M=GBP 5,298.48. Action: Offer early access and loyalty benefits.

Cluster 9: Loyal high-value customers. Customers: 10 (0.2%). Median R=1 days, F=6 orders, M=GBP 1,893.20. Action: Offer early access and loyalty benefits.

Cluster 10: Loyal high-value customers. Customers: 6 (0.1%). Median R=2 days, F=7 orders, M=GBP 2,400.57. Action: Offer early access and loyalty benefits.

DBSCAN radius sensitivity

eps=0.10: 21 clusters; noise=24.9%; non-noise silhouette=-0.356.

eps=0.15: 11 clusters; noise=10.7%; non-noise silhouette=0.109.

eps=0.20: 3 clusters; noise=6.1%; non-noise silhouette=0.247.

eps=0.30: 1 clusters; noise=2.7%; non-noise silhouette=nan.

A small radius can fragment groups and mark many customers as noise; a larger radius can join groups. Choose settings with business review, coverage and stability checks, rather than selecting the largest silhouette alone.